

Osaid Khan

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EDUCATION

Pace University

Master of Science (MS) in Computer Science (GPA - 4.0/4.0)

Jan 2024 - Dec 2025 Dr. A.P.J. Abdul Kalam Technical

University

Noida, India

Bachelor of Technology (B.Tech) in Computer Science

Jul 2016 - Sep 2020

TECHNICAL SKILLS

Languages: Java, JavaScript/TypeScript, Rust, Python

Frameworks: FastAPI, FastMCP, LangGraph, LangChain, LangSmith, PyTorch, React, Sprint Boot

Databases: PostgreSQL, MongoDB, Redis

AWS Services: EC2, RDS, S3, Lambda, CloudWatch, Step Functions, CloudFormation, CloudFront, API Gateway

AI: Claude code, Codex, Pi coding-agent

Others: Google Cloud, Git, CI/CD, Kafka, Node.js

PROFESSIONAL EXPERIENCE

Software Engineer

Phoenix, AZ

Sperse

Feb 2026 - present

- Building the **AI automation layer** for **Sperse**, a multi-tenant SaaS small lenders and fintech businesses use to run the full **lead-to-payment lifecycle** (lead capture, **KBA / credit checks**, contacts and orders, **Stripe / PayPal** payments, **Plaid** bank sync, cash-flow forecasting), turning dozens of admin screens into plain-English conversations and a live AI demo surface for sales
- Designed a **hierarchical multi-agent system** in **LangGraph + FastAPI**: a main orchestrator lazily loads **sub-agent catalogs** (when to use, when not to) and delegates to **40 specialized sub-agents**, each built per request with the user's **JWT, tenant and role context** propagated end-to-end so every backend call respects the caller's permissions
- Compressed **500 backend operations** into **tool-group dispatchers** (lookup, write, activity, etc.) backed by **registry-driven Pydantic validation**, collapsing 60+ REST/OData endpoints per agent into a few typed entry points, **lifting correct tool selection 90%** and cutting **sub-agent context bloat** and tool-call errors
- Wired **LangSmith** end-to-end: every request, delegation and tool call traced; built **trace-driven eval suites** that replay real production conversations against new prompts and model versions, with **real-time monitors** catching agents going off the rails (wrong tool, hallucinated args, runaway loops) before they reach customers
- Added **retrieval-augmented agents** for in-product how-to questions and a **coding agent** that accelerates customer onboarding by **60%** by generating ready-to-run code examples directly from the CRM API docs so customers can build custom UIs and integration flows without leaving the product, protected by **two-stage input safety** (jailbreak regex + **Lakera Guard**) with **SQLite checkpointing** for multi-turn state; separately built a **FastMCP** server exposing **1000+ typed tools** for external AI clients, every call traced in **Langfuse**

AI Researcher

New York City, NY

Pace Artificial Intelligence Lab

Sep 2025 - Dec 2025

- Conducted research on **StyleGAN** and **Stable Diffusion** to generate industry-standard fashion designs using **CUDA**, **PyTorch** and **LORA (Low-rank Adaptation)** on Pace University's **HPC cluster** for model training and fine-tuning
- Utilized **vLLM** for model inference with **Hugging Face** deployment, ensuring secure and efficient model serving
- Conducted workshops on GPU programming with **CUDA**, **Mojo** and building **AI Agents** with **LangChain**, **LangGraph**, and **LangSmith** for +50 students

Software Engineer

Noida, India

Qualitest

Feb 2021 - Dec 2023

- Built an internal support chatbot using **LangChain** and **RAG** (Retrieval-Augmented Generation) with **PostgreSQL** vector store, reducing average ticket resolution time by 20%, enabling natural language queries over documentation
- Designed and developed a cloud-agnostic Storage Service using **Python**, **PostgreSQL**, and **Redis** that seamlessly handles file uploads, downloads, and deletions in **Google Cloud Storage**, **S3**, and **MinIO** through the utilization of expiring signed URLs
- Contributed to the development of an Identity & Access Management platform using **React**, **Spring Boot**, and **AWS**, supporting thousands of users across 30+ applications

- Developed backend workflows using **AWS Step Functions** to handle long-running data import jobs and prevent **AWS Lambda** timeouts

PROJECTS

Dark Factory - Distributed orchestrator for coding agents | *Elixir, TypeScript* github.com/ozzyozbourne/dark-factory

- Turns a single user goal into a **dependency-aware task plan** that is automatically dispatched to a **fleet of AI coding agents running across multiple machines**, with built-in retries and timeout handling
- Built a **fault-tolerant Elixir coordinator** that matches tasks to agents by capability and live workload over a **real-time WebSocket protocol**, with a **CRDT-backed distributed registry** keeping the cluster alive when individual nodes fail